Operating Systems

Program Core

L-T-P-C: 3-1-0-4

1. Course Objectives:

The objective of this course is to provide a solid foundation in Operating System (OS) concepts, including its functions, and interactions with hardware. Students will learn about process management, synchronization, memory allocation, file systems, and system security. The course also covers advanced topics such as deadlocks, virtual memory, and storage management, equipping students with the necessary skills to analyze and optimize modern operating systems for performance, reliability, and security.

2. Syllabus:

Unit – 1 : Introduction - Architecture, Goals & Structures of O.S, Basic functions; Interaction of O.S. & hardware architecture - System calls, Batch, multiprogramming; Multitasking, time sharing, parallel, distributed & real-time O.S.

Unit – 2 : Process Management - Process Concept, Process Scheduling, Operations on Processes, Inter process communication: shared memory systems, message passing, pipes; Threads overview, Multithreading models, CPU scheduling, Scheduling criteria, Scheduling algorithms.

Unit – 3 : Process Synchronization and Deadlock : The Critical-Section Problem, Peterson's Solution, Hardware support for Synchronization, Mutex locks, Semaphores, Classic Problems of Synchronization, Deadlocks - System model, Characterization, Methods for Handling Deadlocks, Deadlock Prevention, Deadlock Avoidance, Deadlock Detection, Recovery from Deadlock.

Unit – 4 : Memory Management: Background, Contiguous memory allocation, Paging, Structure of page table, Swapping, Virtual Memory - Background, Demand Paging, Copy on-write, Page Replacement, Allocation of Frames, Thrashing.

Unit – 5 : Mass storage structure and I/O - Overview of Mass-storage structure, HDD Scheduling, Storage device management, RAID Structure: Reliability via Redundancy, Performance via Parallelism, RAID levels; I/O Hardware, File concept, Access methods, Protection, Directory structure, File-system implementation: File-System structure, File system operations, Directory implementation, Allocation methods.

Unit – 6 : Security & Protection: Security problem, Program threats, System and Network threats, Goals of protection, Principles of protection, Access matrix.

3. Course Outcomes (Unit wise):

At the end of the course, students should have the ability:

I. CO1 : Describe the architecture, structure, and functionalities of operating systems and differentiate various OS types.

II. CO2 : Explain process scheduling, multithreading, and inter-process communication techniques, and evaluate their role in system efficiency.

III. CO3: Implement process synchronization techniques, understand deadlock handling strategies, and apply them to real-world scenarios.

IV. CO4: Analyze memory management schemes such as paging, swapping, and virtual memory, and their impact on system performance.

V. CO5: Understand storage structures, file systems, and RAID technologies, and apply file management techniques for efficient data handling.

VI. CO6: Identify security threats in an OS and apply protection mechanisms to ensure system integrity and secure resource access.

4. Text Books:

a) Abraham Silberschatz, Greg Gagne, Peter B. Galvin, Operating System Concepts, Wiley, 10th Edition, 2018, ISBN: 978-1119320913

b) Andrew Tanenbaum, Modern Operating Systems, Pearson Education India, 4th Edition, 2016, ISBN :

978-9332575776

5. Reference Books:

a) Davie Richard O'Hallaron, and Randal Bryant, Computer Systems: A Programmer's Perspective, Pearson, USA, 3rd Edition, 2015, ISBN: 978-0134092669

b) William Stallings, Operating Systems: Internals and Design Principles, Pearson, 8th Edition, ISBN: 9780133805918

Mapping of Course Outcomes to Programme Outcomes / Program Specific Outcomes

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	2	1	3	2	2	1	1	1
CO2	2	1	4	5	5	3	1	3
CO3	3	2	4	5	5	3	1	3
CO4	3	2	4	5	5	3	1	3
CO5	3	2	4	4	5	3	1	3
CO6	4	2	4	4	4	2	1	2

Course Component

MID ------- 25 %
END ------- 50%
Scheduled Quiz --- 15 % (two quizzez Avg of two)
Class Participation –10 %